

SMART MULTI-MODAL ASSISTIVE OBJECT LOCATOR SYSTEM FOR VISUALLY IMPAIRED USERS USING ESP32, RASPBERRY PI, AND HUSKYLENS

MECHATRONICS SYSTEM INTEGRATION

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SECTION 1

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Abstract

Visually impaired people often have difficulty finding personal items such as wallets, keys, or mobile phones inside their homes. Most existing item-tracking products depend on smartphone screens and visual maps, which makes them hard for blind users to use. This project presents an indoor Smart Assistive Item Locator System that helps visually impaired users find misplaced objects using simple and intuitive feedback methods.

The proposed system employs a centralized indoor architecture consisting of two ESP32 microcontrollers and a Raspberry Pi acting as a home-based control unit. A wearable bracelet equipped with an ESP32 allows user interaction via push buttons and delivers guidance using a vibration motor. An object-mounted ESP32 activates an acoustic buzzer to indicate the location of the selected item. Communication between system components is achieved through WiFi within the home network, enabling reliable indoor operation and centralized decision-making.

To enhance object confirmation and directional guidance, a HuskyLens vision sensor is integrated into the wearable unit. When the target object is visually detected, tactile feedback is provided through vibration, allowing the user to orient themselves toward the object without relying solely on sound. A single 3.7 V Li-ion battery powers the wearable system, with a TP4056 charging module and a DC-DC boost converter ensuring stable operation.

The proposed system emphasizes affordability, accessibility, and user-centered design, offering an effective indoor assistive solution that improves independence and usability for visually impaired individuals.

Keywords: Visually Impaired Assistance Indoor Item Locator Assistive Technology
ESP32 Raspberry Pi HuskyLens Tactile Feedback Smart Wearable System

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1. Introduction

Visually impaired individuals face continuous challenges in performing daily activities independently, particularly when locating personal belongings within indoor environments. Common items such as wallets, keys, and mobile phones are often misplaced, and without visual cues, identifying their location can be time-consuming and mentally exhausting. This limitation not only reduces efficiency but also affects the user's confidence and independence in managing daily routines.

Existing item locator solutions, such as Bluetooth trackers and smart tags, are widely available in the consumer market. However, most of these systems rely heavily on smartphone-based visual interfaces, maps, or screen notifications, making them unsuitable for blind or visually impaired users. While some devices provide audible alerts, they lack directional guidance, tactile feedback, and intelligent confirmation of the target object. As a result, visually impaired users may still struggle to accurately locate and physically retrieve the desired item, especially in cluttered indoor environments.

Recent advancements in embedded systems, wireless communication, and artificial intelligence have enabled the development of assistive technologies that are more accessible and user-centered. Microcontrollers such as the ESP32 offer low-power wireless communication and real-time input-output control, while microprocessors like the Raspberry Pi provide centralized processing and system coordination. In addition, AI-based vision sensors such as HuskyLens enable real-time object recognition without requiring complex external computation or visual displays.

This project proposes a Smart Multi-Modal Assistive Object Locator System designed specifically for visually impaired users in indoor environments. The system integrates auditory, tactile, and visual intelligence to assist users in locating misplaced personal items effectively. A wearable bracelet equipped with an ESP32 microcontroller serves as the primary user interface, allowing users to initiate item searches using simple push buttons. Target objects such as wallets, keys, or mobile phones are equipped with ESP32-based locator units that activate audible buzzers upon receiving search commands.

A Raspberry Pi functions as a centralized home control unit, coordinating communication between the wearable bracelet and the object locator units through WiFi. This centralized

architecture enhances system reliability and simplifies indoor operation by enabling structured communication and decision-making. To further improve accuracy and usability, a HuskyLens vision sensor is integrated into the wearable bracelet. Once the user is near the sound source, the HuskyLens detects and recognizes the target object, providing tactile feedback through a vibration motor to guide the user toward the precise location of the item.

Compared to existing item locator systems, the proposed solution emphasizes accessibility, multimodal feedback, and independence from visual interfaces. By combining audio alerts, tactile vibration, and AI-based object recognition within a wearable system, this project aims to improve the independence, efficiency, and overall quality of life for visually impaired individuals. From a Mechatronics System Integration perspective, the project demonstrates effective integration of microcontrollers, microprocessors, wireless communication, and intelligent sensing within a unified assistive system.

Table 1 Shows the Comparison of Existing Item Locator Systems

Ref / Year	Title	Advantages	Disadvantages	Compared to Proposed System
	Apple AirTag	High location accuracy, wide ecosystem support	Visual-only interface, no tactile guidance, requires smartphone	Proposed system provides vibration and audio guidance without visual dependency
	Tile Bluetooth Tracker	Low cost, audible alert	Limited range, smartphone-dependent, no intelligent guidance	Proposed system offers centralized control and vision-assisted confirmation
	Bluetooth-Based Indoor Object Finder	Simple implementation, buzzer alert	No directional or tactile feedback, not designed for blind users	Proposed system integrates tactile vibration and AI-based object detection
Proposed	Smart Assistive Item Locator System	Multimodal feedback, vision-assisted guidance, affordable, accessible	Limited to indoor environments	Designed specifically for visually impaired users

2. Methodology

2.1 Project Development

The overall system architecture is illustrated using a block diagram that represents the interaction between input devices, processing units, communication modules, and output mechanisms. The proposed project consists of three main subsystems: the wearable bracelet unit, the object locator unit, and the home central control unit.

System 1: Wearable Bracelet Unit

This subsystem is built around an ESP32 microcontroller and serves as the primary user interface. It includes three push buttons corresponding to different personal items, a vibration motor for tactile feedback, and a HuskyLens vision sensor for object detection. User input is captured through button presses, and guidance is provided through vibration when the target object is detected.

System 2: Object Locator Unit

The object locator unit consists of an ESP32 microcontroller attached to the target item. Three buzzers are connected to the microcontroller to provide audible cues indicating the object's location. Upon receiving commands from the Raspberry Pi, the appropriate buzzer is activated.

System 3: Home Central Control Unit

The Raspberry Pi functions as the central control unit and coordinates communication between the wearable bracelet and the object locator units via WiFi. It manages system logic, processes search requests, and determines when to activate audio or tactile feedback. The centralized architecture ensures reliable indoor operation and simplifies system management.

A flowchart illustrates the operational sequence of the system, beginning with user input, followed by communication with the Raspberry Pi, object detection, and feedback activation.

The block diagram of the overall system is shown in Fig. 1, where the wearable bracelet communicates wirelessly with both the object locator and the Raspberry Pi. User input is provided through push buttons on the bracelet, while feedback is delivered through a vibration motor. The HuskyLens vision sensor is interfaced with the bracelet-ESP32 to enable visual detection of the target object.

System 1, illustrated in Fig. 2, consists of an ESP32 microcontroller connected to three push buttons, a vibration motor, and the HuskyLens module. Each button corresponds to a specific object (wallet, keys, or phone). Upon button activation, a request is sent to the Raspberry Pi, which coordinates the object search process.

System 2, shown in Fig. 3, represents the object locator unit attached to the target item. This subsystem uses an ESP32 microcontroller connected to three buzzers, each representing a distinct object. Upon receiving a command from the Raspberry Pi, the corresponding buzzer is activated to provide an audible cue indicating the object's location.

System 3 comprises the Raspberry Pi, which functions as the central control unit. It manages wireless communication, processes object detection status, and determines when to activate vibration or buzzer feedback. The flowchart presented in Fig. 4 illustrates the operational logic of the entire system, from button press detection to object localization and guidance feedback.

2.2 Schematic Diagram and Prototype Setup

The schematic diagram shows how all the parts are connected in each part of the system. The ESP32 in the wearable bracelet is connected to three push buttons, each with a pull-down resistor to make sure the button signals are stable. A small circuit with a transistor is used to turn the vibration motor on and off without drawing too much current from the ESP32. A diode is placed across the motor to protect the circuit from sudden voltage spikes.

The ESP32 on the object controls three buzzers, each connected to its own pin through a resistor to limit the current. The ESP32 modules and the Raspberry Pi communicate wirelessly through WiFi, so there is no need for wires between the parts.

The wearable unit is powered by a single 3.7 V Li-ion battery. A TP4056 charging module is used to charge the battery and provide power to the ESP32, vibration motor, and a DC-DC boost converter. The boost converter gives a steady 5 V needed for the HuskyLens. The Raspberry Pi uses a 5 V power supply and stays on all the time as the home hub.

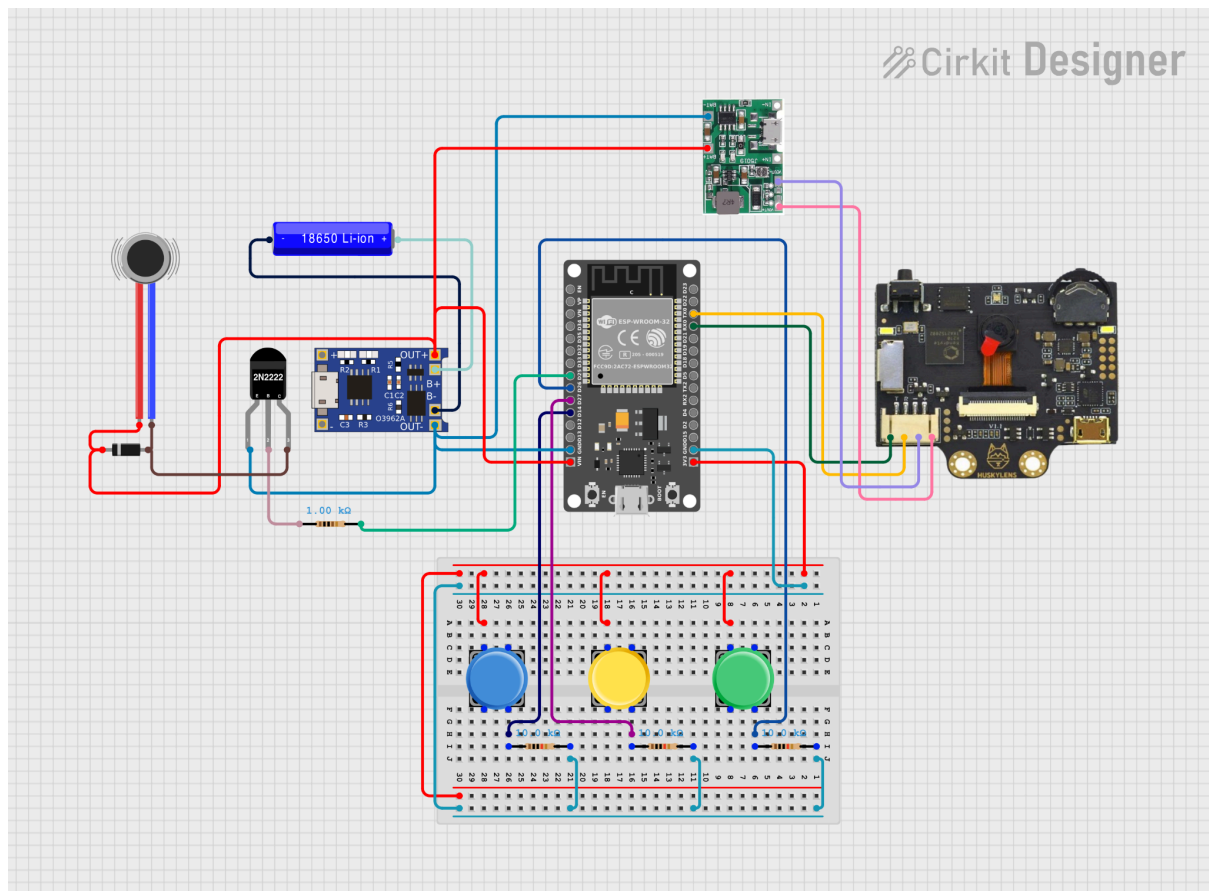


Figure 2.2.0 Electrical design for ESP32-Bracelet

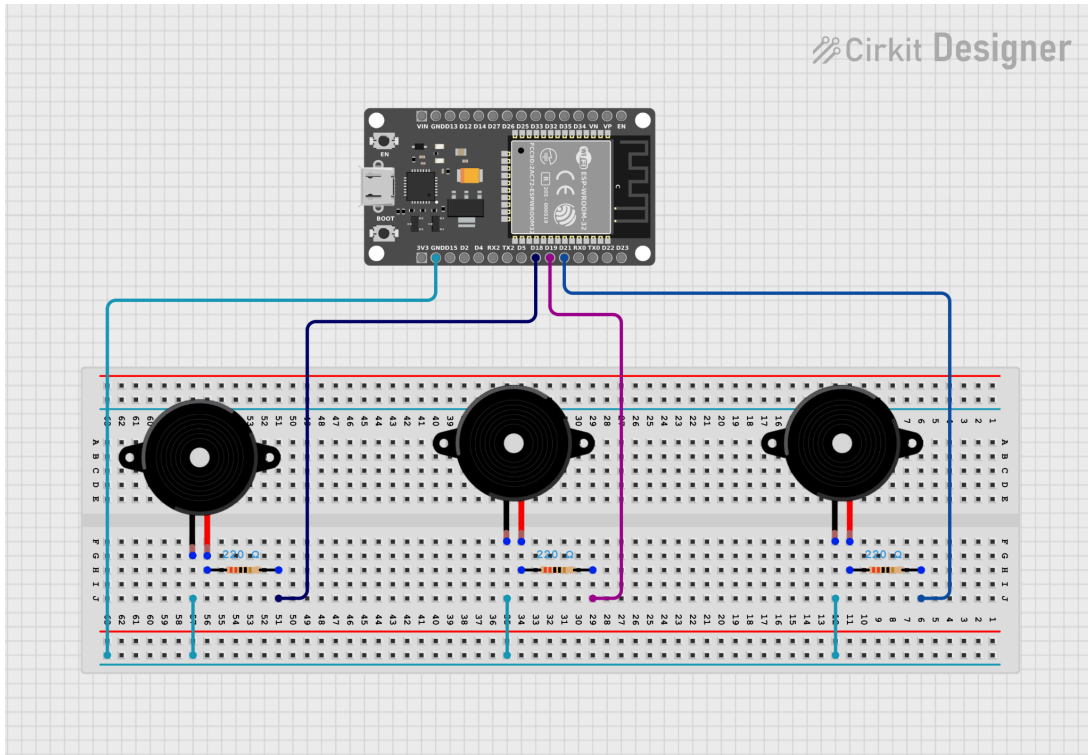


Figure 2.2.1 Electrical design for ESP32-Object Unit

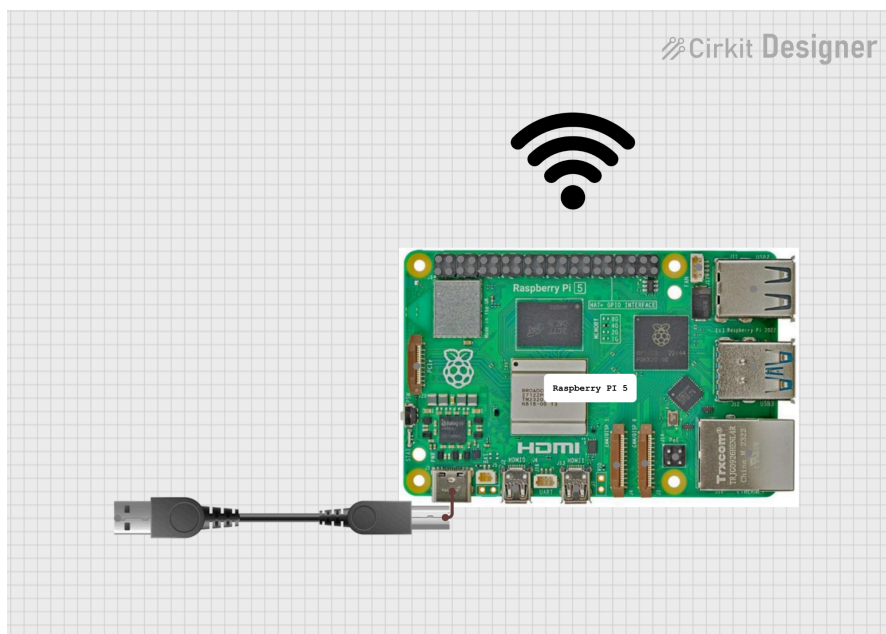


Figure 2.2.2 Smart Multi-Modal Assistive Object Locator System

2.3 Proposed System Architecture and Operational Workflow

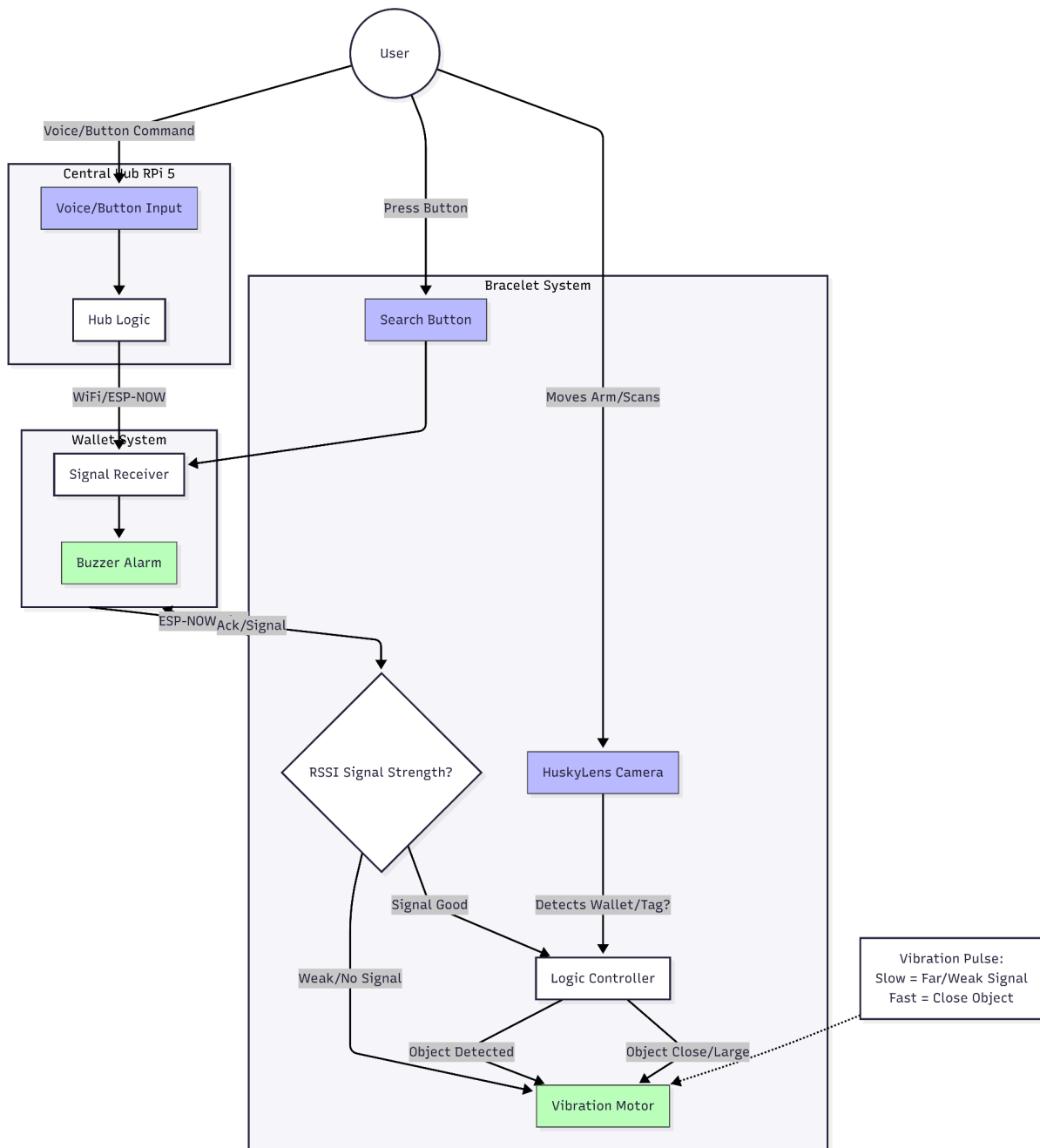


Figure 1.0

Figure 1.0 (referring to your flowchart) illustrates the complete operational logic of the Assistive Blind Locator System. The architecture is divided into three primary subsystems: the **Bracelet System** (Wearable Master), the **Wallet System** (Target Device), and the **Central**

Hub (Stationary Backup). These systems interact through wireless protocols (ESP-NOW/Wi-Fi) to provide redundancy and multi-sensory feedback to the user.

The system workflow is categorized into three distinct functional cases as detailed below:

1. Active Audio Search (Remote Location)

This workflow allows the user to locate the wallet audibly when it is out of immediate reach but within the house. As depicted in the "User" interaction paths (Top Center), there are two methods to trigger this alarm:

- **Primary Method (Bracelet):** The user presses the physical **Search Button** on the Bracelet System. This sends a direct signal to the **Signal Receiver** on the Wallet System, instantly triggering the **Buzzer Alarm**.
- **Secondary Method (Central Hub Backup):** If the bracelet is disconnected or unavailable, the user can issue a **Voice/Button Command** to the **Central Hub RPi 5**. The **Hub Logic** processes this request and transmits a signal via Wi-Fi/ESP-NOW to the Wallet System, activating the same **Buzzer Alarm**.

2. Passive Tethering (Anti-Loss Monitoring)

To prevent the user from losing the wallet initially, the system runs a continuous background monitoring loop shown in the lower-left logic path:

- **Signal Acquisition:** The Bracelet System continuously exchanges "Ack/Signal" packets with the Wallet System using the low-latency ESP-NOW protocol.
- **RSSI Evaluation:** The **Logic Controller** analyzes the **RSSI Signal Strength** (Received Signal Strength Indicator) of these packets.
- **Loss Alert:** If the signal is determined to be "Weak/No Signal" (indicating the user has walked away from the wallet), the logic triggers the **Vibration Motor** on the bracelet. This acts as a preventative "virtual leash," alerting the user immediately if they leave the wallet behind.

3. Visual Haptic Guidance (Precision Pinpointing)

Once the user is in close proximity to the wallet (guided by the buzzer or memory), the system switches to precision guidance to help the user physically grasp the item:

- **Visual Recognition:** As the user **Moves Arm/Scans** the area, the **HuskyLens Camera** integrated into the bracelet processes the video feed in real-time.
- **Object Detection Logic:** The camera queries "Detects Wallet/Tag?". Upon positive identification of the wallet (or its specific marker), the **Logic Controller** engages the **Vibration Motor**.
- **Variable Feedback Loop:** As noted in the diagram's legend (*Vibration Pulse*), the system utilizes a variable haptic language:
 - **Slow Pulse:** Indicates the object is detected but is distant or small in the frame.
 - **Fast Pulse:** Indicates the object is close or large in the frame, signaling the user to reach out.

Summary of System Logic

This architecture ensures high reliability by combining **redundancy** (RPi Hub acts as a fail-safe for the Bracelet), **prevention** (RSSI Tethering prevents loss before it happens), and **accessibility** (translating visual data from the Camera into haptic vibrations), thereby creating a complete safety net for the visually impaired user.

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Certificate of Originality and Authenticity

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We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We, therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us**.

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